

IV B.Tech. I Semester MODEL QUESTION PAPER

DEEP LEARNING TECHNIQUES

(for CSBS)

Time: 3 Hrs.

Max. Marks: 70 M

Answer Question No.1 compulsorily

Answer **ONE** Question from **EACH UNIT**

Assume suitable data if necessary

10 x 2 = 20 Marks

			CO	KL	M
1.	a).	What is McCulloch–Pitts model?	1	1	2
	b).	What is a perceptron?	1	1	2
	c).	What is gradient descent?	2	1	2
	d).	What is regularization?	2	1	2
	e).	Define dropout.	3	1	2
	f).	What is pooling?	3	1	2
	g).	Define Long Short-Term Memory (LSTM).	4	1	2
	h).	What is Gibbs sampling?	4	1	2
	i).	What is a Transformer model?	5	1	2
	j).	Mention any two applications of deep learning in speech processing.	5	2	2

5 x 10 = 50 Marks

		UNIT-1	CO	KL	M
2.	a).	Explain biological neuron vs artificial neuron.	1	2	5
	b).	Explain convergence theorem of perceptron.	1	2	5
		OR			
3.	a).	Explain thresholding logic with examples.	1	2	5
	b).	Explain perceptron learning algorithm.	1	2	5
		UNIT-2			
4.		Explain Multilayer Perceptron architecture and back propagation algorithm with diagram.	2	2	10
		OR			
5.	a).	Explain empirical risk minimization and its role in model training.	2	2	5
	b).	Describe the concept and procedure of greedy layer-wise training.	2	2	5
		UNIT-3			
6.	a).	Apply Adagrad optimizer for weight update with example.	3	3	5
	b).	Apply pooling operation (max/avg) with example.	3	3	5
		OR			

7.	a).	Apply Adam optimizer and compute weight updates.	3	3	5
	b).	Demonstrate batch normalization on sample data.	3	3	5
		UNIT-4			
8.	a).	Demonstrate bidirectional RNN working.	4	3	5
	b).	Apply CNN architecture (LeNet/AlexNet) and compute layer-wise output dimensions.	4	3	5
		OR			
9.	a).	Apply LSTM gate equations to show how vanishing gradient problem is overcome.	4	3	5
	b).	Demonstrate gradient computation in RBM.	4	3	5
		UNIT-5			
10.	a).	Describe the architecture of the Transformer model.	5	2	5
	b).	Explain the use of deep learning in speech processing.	5	2	5
		OR			
11.	a).	Explain the concept of Variational Autoencoders.	5	2	5
	b).	Explain applications of deep learning in computer vision.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks



Course Code: B23HS4102					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
PRODUCTION & OPERATIONS MANAGEMENT					
(Common to CIC & CSBS)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define Production & Operations Management in the context of a modern enterprise.	1	1	2
	b).	State the difference between the 'Controllable' and 'Uncontrollable' variables influencing an operations system	1	1	2
	c).	State two distinct characteristics of a Job Shop Production System.	2	1	2
	d).	Define 'Floor Plan Mapping' in facility design.	2	1	2
	e).	Define Supply Chain Management	3	1	2
	f).	What is meant by lead time in inventory management?	3	1	2
	g).	What is Kaizen?	4	1	2
	h).	What is Statistical Process Control	4	1	2
	i).	What is PERT.	5	1	2
	j).	Define critical path.	5	1	2
5 x 10 =50Marks					
		UNIT-1			
2.		Explain the Variables influencing the Production & Operations Management.	1	3	10
		OR			
3.		Discuss the evolution of Operations Management with historical perspectives.	1	3	10
		UNIT-2			
4.		Discuss the factors influencing the Site selection Criteria	2	3	10
		OR			
5.		Distinguish between mass, batch and continuous production systems.	2	3	10
		UNIT-3			
6.	a).	Discuss the ABC analysis technique in inventory control	3	3	5

	b).	Explain the Just-In-Time (JIT) inventory system	3	3	5																		
		OR																					
7.		Discuss the qualitative and quantitative demand forecasting techniques	3	3	10																		
		UNIT-4																					
8.	a).	Describe the benchmarking process and list its advantages	4	3	5																		
	b).	Explain the Six Sigma approach to quality improvement.	4	3	5																		
		OR																					
9.	a).	Discuss the ISO series of quality standards	4	3	5																		
	b).	Describe the principles of Total Quality Management	4	3	5																		
		UNIT-5																					
10.	a).	Discuss the techniques of capacity planning.	5	3	5																		
	b).	Describe the strategic production planning process	5	3	5																		
		OR																					
11.		Draw the network and find critical path, the minimum time to complete project and total float of each activity.																					
		<table><tr><td>ACTIVITY</td><td>1-2</td><td>1-3</td><td>2-3</td><td>2-4</td><td>3-4</td><td>5</td><td>4-5</td><td>3</td></tr><tr><td>TIME</td><td>20</td><td>25</td><td>10</td><td>12</td><td>6</td><td></td><td>10</td><td></td></tr></table>	ACTIVITY	1-2	1-3	2-3	2-4	3-4	5	4-5	3	TIME	20	25	10	12	6		10				10
ACTIVITY	1-2	1-3	2-3	2-4	3-4	5	4-5	3															
TIME	20	25	10	12	6		10																

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Estd. 1980

Course Code: B23CS4104					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
AUGUMENTED REALITY & VIRTUAL REALITY					
(Common to CSE & CSBS)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Build the history of augmented reality explain in briefly.	1	3	2
	b).	Apply characteristics of tracking technology.	1	3	2
	c).	Create a quick explanation of Outdoor Tracking.	2	3	2
	d).	Select the Scene Graphs and explain them briefly.	2	3	2
	e).	Make the use of human physiology and perception in VR.	3	3	2
	f).	Use visual distortions in virtual reality (VR) temporarily.	3	3	2
	g).	Select the consequences for virtual reality (VR) briefly.	4	3	2
	h).	Choose how to repair optical distortions briefly.	4	3	2
	i).	Select The Vestibular System: A Brief Explanation.	5	3	2
	j).	Choose the Auditory Rendering and briefly explain it.	5	3	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Make use of augmented reality (AR), provide instances of AR applications, and talk about adjacent topics like virtual reality (VR) and mixed reality.	1	3	5
	b).	Apply visual perception to augmented reality (AR) and quickly define multimodal displays.	1	3	5
		OR			
3.	a).	Chose the requirements and characteristics of AR displays, as well as the spatial display model and visual displays.	1	3	5
	b).	Make the use of coordinate systems used in AR tracking, as well as the differences between stationary tracking systems and mobile sensors.	1	3	5
		UNIT-2			
4.	a).	Choose "How does AR marker track work?" Describe briefly the benefits of tracking natural features.	2	3	5
	b).	Make advantage of the various AR interaction modalities, and briefly discuss the limitations in outside AR tracking.	2	3	5
		OR			
5.	a).	Choose one of the following interaction modalities: haptic interaction in augmented reality, virtual user interfaces on actual surfaces and augmented paper, output, input, and tangible interfaces, and explain it	2	3	5

		succinctly.			
	b).	Make use of the scene graph function in augmented reality apps and explain it briefly.	2	3	5
		UNIT-3			
6.	a).	Choose virtual reality (VR) and provide a brief overview of human physiology and perception in virtual reality.	3	3	5
	b).	It explains how to use axis-angle rotations, geometric models, VR viewing transformations, and light and optics, such as lenses, aberrations, the human eye, cameras and displays.	3	3	5
		OR			
7.	a).	Choose the virtual reality (VR) and its main features. How does human perception affect VR design?	3	3	5
	b).	Make use of lenses and optical imperfections. VR. How do cameras and screens work in VR systems?	3	3	5
		UNIT-4			
8.	a).	Identify the routes that lead from photoreceptors to the cortex of the eye. How do the cornea and photoreceptors of the human eye process light?	4	3	5
	b).	Create eye movements and investigate how they influence VR design, depth, motion, and color perception.	4	3	5
		OR			
9.	a).	Develop the visual cortex's role in perception. How do photoreceptors, such as rods and cones, function? How does virtual reality utilize depth perception?	4	3	5
	b).	Select visual rendering techniques such as ray tracing, shading, rasterization, fixing optical aberrations and latency in virtual reality, and creating immersive images and films for VR.	4	3	5
		UNIT-5			
10.	a).	Make use of the visual pathways that connect photoreceptors and the visual cortex. How does light pass from the cornea to the photoreceptors?	5	3	5
	b).	Use eye movements, types, and implications for VR design, and in-depth perception of VR cues (binocular disparity, etc.) to describe it.	5	3	5
		OR			
11.	a).	Make use of the way motion is perceived. How VR uses motion perception, color perception, the influence of color vision in VR, and rendering in VR: ray tracing versus rasterization to designate it.	5	3	5
	b).	Select the distortion correction option. How and why is it implemented in virtual reality? Outline latency reduction strategies and methods for improving virtual reality.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23CS4102					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
SOFTWARE ARCHITECTURE & DESIGN PATTERNS					
(Common to CSE,AIDS, CSBS, CSD)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Explain the role of Model–View–Controller (MVC) in design patterns.	1	2	2
	b).	Differentiate between cohesion and coupling.	1	2	2
	c).	Explain how actors are identified in use-case analysis.	2	2	2
	d).	Explain the concept of conceptual classes in system analysis.	2	2	2
	e).	Summarize the key idea behind the Adapter pattern.	3	2	2
	f).	Describe how the Proxy pattern controls access to another object.	3	2	2
	g).	Explain the role of the Model component in MVC architecture.	4	2	2
	h).	Explain how the undo operation can be supported in an interactive system.	4	2	2
	i).	Explain the concept of a distributed object system.	5	2	2
	j).	Explain the roles of client and server in a distributed application.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Describe the essential elements used to document a design pattern.	1	2	5
	b).	Describe how design patterns help solve common design problems in software development.	1	2	5
		OR			
3.	a).	Describe the key concepts of object-oriented design.	1	2	5
	b).	Describe the drawbacks of the object-oriented paradigm.	1	2	5
		UNIT-2			
4.	a).	Use requirement gathering techniques to prepare a functional requirement specification for a student attendance management system.	2	3	5
	b).	Demonstrate how use-case analysis can be used to represent the functional requirements of a bank ATM system.	2	3	5
		OR			
5.	a).	Illustrate the process of identifying conceptual classes and relationships from a given problem description.	2	3	5
	b).	Apply domain knowledge to refine the analysis model of an e-commerce order processing system.	2	3	5

		UNIT-3			
6.	a).	Demonstrate how the Bridge pattern can be used to separate device abstraction from its implementation in a remote-control system.	3	3	5
	b).	Illustrate the use of the Decorator pattern to dynamically add additional features to a text editor component.	3	3	5
		OR			
7.	a).	Apply the Facade pattern to simplify the interaction between multiple subsystems in a banking application.	3	3	5
	b).	Demonstrate how the Flyweight pattern can be used to optimize memory usage in a large document editor with many repeated characters.	3	3	5
		UNIT-4			
8.	a).	Apply the MVC architectural pattern to design an interactive drawing application, identifying the model, view, and controller components.	4	3	5
	b).	Apply MVC principles to design the subsystems of an interactive online shopping interface.	4	3	5
		OR			
9.	a).	Demonstrate how a new feature can be added to an MVC-based system without modifying the entire architecture.	4	3	5
	b).	Illustrate how pattern-based solutions improve extensibility in an MVC-based interactive system.	4		5
		UNIT-5			
10.	a).	Demonstrate how Java Remote Method Invocation (RMI) can be used to enable communication between distributed objects in a networked application.	5	3	5
	b).	Demonstrate how RESTful web services can be used to provide data access in a student information system.	5	3	5
		OR			
11.	a).	Illustrate the process of implementing an object-oriented system on the web for an online shopping application.	5	3	5
	b).	Illustrate how an Enterprise Service Bus (ESB) can be used to integrate different enterprise applications in a distributed environment.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

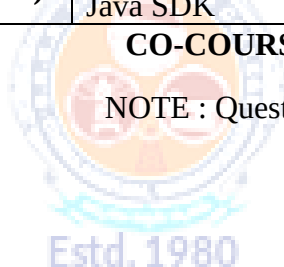
NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code:B23IT4104					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
BLOCKCHAIN TECHNOLOGY & APPLICATIONS					
(Common to IT, AIDS, CSBS)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			C O	KL	M
1.	a).	Define Blockchain and list its components.	1	1	2
	b).	What is decentralization in blockchain?	1	2	2
	c).	Define Smart Contract.	2	1	2
	d).	What are Oracles in blockchain?	2	2	2
	e).	What is a Private Blockchain?	3	1	2
	f).	Define Byzantine Fault Tolerance.	3	2	2
	g).	What is Blockchain scalability issue?	4	2	2
	h).	List any two applications of blockchain	4	2	2
	i).	What is Hyperledger Fabric?	5	1	2
	j).	List any two blockchain case study domains.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the origin and evolution of blockchain technology	1	2	5
	b).	Describe different types of blockchain and consensus mechanisms	1	2	5
		OR			
3.	a).	Explain the structure of a block in a blockchain with a neat diagram	1	2	5
	b).	Discuss cryptocurrency basics and types with suitable examples.	1	2	5
		UNIT-2			
4.		Explain Public Blockchain systems and the working of Bitcoin and Ethereum blockchain	2	3	10
		OR			
5.		Explain Smart Contracts, their characteristics, types, and role of oracles in blockchain	2	3	10
		UNIT-3			
6.	a).	Explain the architecture and characteristics of Private Blockchain systems	3	2	5

	b).	Describe Multichain and Byzantine Fault Tolerance in permissioned blockchain	3	3	5
		OR			
7.	a).	Explain Consortium Blockchain and its need in industry	3	2	5
	b).	Discuss Hyperledger, Ripple, and Corda platforms	3	2	5
		UNIT-4			
8.		Explain security and privacy challenges in blockchain along with identity management and smart contract security	4	3	10
		OR			
9.		Discuss various applications of blockchain in different domains and explain its limitations	4	2	10
		UNIT-5			
10.	a).	Explain blockchain case studies in retail and banking sectors	5	3	5
	b).	Describe blockchain implementation using Python and relevant packages	5	2	5
		OR			
11.	a).	Explain components of Hyperledger Fabric architecture.	5	2	5
	b).	Discuss chaincode and blockchain application development using Fabric Java SDK	5	3	5

CO-COURSE OUTCOME KL-KNOWLEDGE LEVEL M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks



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Course Code: B23CB4102					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
EMBEDDED SYSTEMS					
(for CSBS)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define an embedded system and give two examples.	1	1	2
	b).	List any four characteristics of embedded systems.	1	1	2
	c).	Define a typical embedded system and list its core components.	2	1	2
	d).	Differentiate between General Purpose Processors (GPP) and Domain Specific Processors (DSP) used in embedded systems.	2	2	2
	e).	What is the purpose of a watchdog timer in an embedded system?	3	2	2
	f).	Define brown-out protection circuit and explain its importance in embedded firmware.	3	1	2
	g).	Define multitasking in an RTOS-based embedded system.	4	1	2
	h).	Differentiate between a process and a thread.	4	2	2
	i).	What is shared memory in task communication?	5	1	2
	j).	Define task synchronization and mention any two synchronization techniques used in RTOS.	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Compare embedded systems with general computing systems in terms of functionality, performance, and applications.	1	2	5
	b).	Explain the history and evolution of embedded systems. Discuss the major milestones in embedded system development.	1	2	5
		OR			
3.	a).	Describe the classification of embedded systems based on performance and functional requirements with suitable examples.	1	2	5
	b).	Discuss the major application areas, purpose, characteristics, and quality attributes of embedded systems.	1	2	5
		UNIT-2			
4.	a).	Explain the architecture of a typical embedded system. Discuss the role of processors, memory, sensors, actuators, and communication interfaces in the system.	2	2	5
	b).	Compare ASICs, PLDs, and Commercial Off-The-Shelf (COTS)	2	2	5

		components used in embedded systems. Discuss their advantages, disadvantages, and applications.			
		OR			
5.	a).	Describe the different types of memory used in embedded systems, including ROM, RAM, and interface-based memory. Explain the concept of memory shadowing and memory selection criteria.	2	2	5
	b).	Explain onboard and external communication interfaces used in embedded systems. Discuss the importance of sensors and actuators in interfacing the embedded system with the physical environment.	2	2	5
		UNIT-3			
6.	a).	Explain the functions of the reset circuit, brown-out protection circuit, and oscillator unit in an embedded system.	3	2	5
	b).	Apply a Real Time Clock (RTC) to a real time scenario and explain its operation, How does it differ from the system clock?	3	3	5
		OR			
7.	a).	Explain the working principle of a watchdog timer. Discuss how it improves the reliability of embedded systems.	3	2	5
	b).	A smart home device frequently resets due to voltage fluctuations. Apply the concept of brown-out protection and watchdog timers to improve the reliability of the system.	3	3	5
		UNIT-4			
8.	a).	Explain the basic functions of an operating system in embedded systems. Discuss the importance of RTOS in embedded applications.	4	2	5
	b).	Describe the different types of operating systems used in embedded systems with suitable examples.	4	2	5
		OR			
9.	a).	Explain the concepts of tasks, processes, and threads in an RTOS-based embedded system. Compare their characteristics and applications.	4	2	5
	b).	Discuss multiprocessing, multitasking, and task scheduling in embedded systems. Explain different scheduling techniques used in RTOS.	4	2	5
		UNIT-5			
10.	a).	Explain different task communication methods used in RTOS, including shared memory, message passing, Remote Procedure Call (RPC), and sockets.	5	1	5
	b).	Discuss the issues involved in task communication and synchronization in embedded systems. How can these issues be resolved?	5	2	5
		OR			
11.	a).	Explain various task synchronization techniques used in RTOS, such as semaphores, mutexes, and event flags, with suitable examples.	5	1	5

	b).	Describe the role of device drivers in embedded systems. Discuss the important factors and methods used to choose an appropriate RTOS for an embedded application.	5	2	5
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CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks



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Course Code: B23CS4107					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
AGILE METHODOLOGIES					
(Common to CSE, AIML, AIDS, CSBS, IT)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a)	What is meant by Agile manifesto	1	1	2
	b)	Define task board	1	1	2
	c)	Define self organizing team	2	1	2
	d)	Suggest appropriate agile communication principles for a software development team that is struggling with communication gaps and delayed deliveries in Ebook Reader project.	2	3	2
	e)	Mention the three main roles on a Scrum project.	3	1	2
	f)	How is Scrum both incremental and iterative?	3	2	2
	g)	Define pair programming	4	1	2
	h)	What is feedback loops and why do XP teams create them	4	2	2
	i)	Define Shuhari	5	1	2
	j)	What are the types of waste that create constraints in the workflow?	5	1	2
5 x 10 = 50 Marks					
UNIT-1					
2.	a)	Explain the core values of Agile manifesto and describe how they differ from traditional software development approaches	1	2	5
	b)	Analyze how a “fractured perspective” leads to project failure. How does Agile thinking help resolve this issue?	1	3	5
OR					
3.	a)	Explain the idea of “Principles over Practices” in Agile. Why is focusing only on practices without understanding underlying principles often ineffective	1	2	5
	b)	Illustrate with an example how “Responding to Change over Following a Plan” benefits a software project.	1	3	5
UNIT-2					
4.		Explain Agile principles and their importance in software development.	2	2	10
OR					
5.	a)	Explain how effective communication contributes to better project delivery in software development.	2	2	5
	b)	Explain how continuous improvement contributes to better project and team performance in Agile	2	2	5
UNIT-3					
6.	a)	Explain the Scrum framework. Discuss the roles of Product Owner, Scrum master, and development team	3	2	5

	b)	Design a complete sprint planning process for a mobile application development project. Apply user stories, story points. Velocity estimation and task board management to create a comprehensive sprint plan.	3	3	5
		OR			
7.	a)	Explain the significance of Scrum values. Explain why Scrum practices alone are insufficient without adherence to these values.	3	2	5
	b)	A sprint fails because the Product Owner frequently changes priorities during sprint. Use Scrum planning rules to analyze the product owner support for sprint success without disrupting execution.	3	3	5
		UNIT-4			
8.		Explain the XP values and their role in shaping an effective development mindset	4	2	10
		OR			
9.	a)	Illustrate how continuous integration helps in maintaining software quality in XP	4	2	5
	b)	Explain how XP practices help teams embrace changing requirements effectively.	4	2	5
		UNIT-5			
10.	a)	Explain Lean thinking and identify different types of waste in software development	5	2	5
	b)	Analyse how limiting Work in Progress (WIP) improves flow in a Kanban system	5	3	5
		OR			
11.	a)	Explain Kanban principles to manage workflow in a software maintenance project	5	2	5
	b)	Evaluate the role of Agile Coach in helping organizations adopt agile practices successfully	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23CI4109					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
METAVERSE					
(Common to CIC & CSBS)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Differentiate between VR and AR.	1	2	2
	b).	Apply the concept of Blockchain in virtual transactions	1	3	2
	c).	Explain the role of sensors in VR systems.	2	2	2
	d).	Apply VR technology in training simulations.	2	3	2
	e).	How are NFTs different from cryptocurrencies?	3	2	2
	f).	Name one blockchain platform.	3	3	2
	g).	Explain the concept of digital goods with examples	4	2	2
	h).	Explain how digital identity works in the Metaverse	4	2	2
	i).	State the purpose of WebXR in Metaverse applications	5	2	2
	j).	What is the purpose of interoperability standards?	5	2	2
5 x 10 = 50 Marks					
UNIT-1					
2.	a).	Explain the applications of the Metaverse in gaming and entertainment.	1	3	5
	b).	Analyze the challenges faced in the development of the Metaverse.	1	4	5
OR					
3.	a).	Explain the role of the Metaverse in healthcare with examples.	1	3	5
	b).	Compare VR, AR, and MR technologies in terms of features and applications	1	4	5
UNIT-2					
4.	a).	Describe hardware and software requirements for VR/AR.	2	2	5
	b).	Analyze the role of Metaverse platforms like Oculus and Microsoft Mesh	2	4	5
OR					
5.	a).	Explain interaction techniques used in VR systems.	2	2	5
	b).	Evaluate challenges in designing VR/AR systems.	2	4	5
UNIT-3					
6.	a).	Explain how Blockchain technology works.	3	2	5
	b).	Analyze the role of decentralization in the Metaverse ecosystem	3	4	5

		OR			
7.	a).	Describe the working of Smart Contracts in blockchain systems	3	2	5
	b).	Analyze security and privacy issues in Blockchain systems.	3	4	5
		UNIT-4			
8.	a).	Explain the Play-to-Earn (P2E) model with suitable examples.	4	2	5
	b).	Discuss how Decentralized Finance (DeFi) supports the Metaverse economy.	4	3	5
		OR			
9.	a).	Explain the concept of tokenomics and its key components.	4	2	5
	b).	Analyze ethical issues such as privacy and security in the Metaverse.	4	4	5
		UNIT-5			
10.	a).	Explain the features and applications of Unity in Metaverse development.	5	2	5
	b).	Analyze the advantages and limitations of WebXR	5	4	5
		OR			
11.	a).	Explain the role of Artificial Intelligence in enhancing user experience in the Metaverse	5	2	5
	b).	Analyze how AI and cloud computing work together in Metaverse platforms.	5	4	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Estd. 1980

Course Code: B23CS4109					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
COMPUTER VISION					
(Common to CSE, CSD, IT & CSBS)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What is Radiometry? Why is it important in computer vision?	1	2	2
	b).	Define surface color.	1	1	2
	c).	What is the Fourier Transform?	2	1	2
	d).	What is meant by edge detection?	2	1	2
	e).	Define stereopsis.	3	1	2
	f).	What are Gestalt principles?	3	2	2
	g).	How are lines fitted using the Hough Transform?	4	2	2
	h).	What is Kalman Filtering?	4	1	2
	i).	What are camera parameters?	5	1	2
	j).	What is pose clustering?	5	1	2
5 x 10 = 50 Marks					
		UNIT-I			
2.	a).	Explain the Pinhole Camera Model with a neat diagram and derive the projection equation.	1	2	5
	b).	Explain Photometric Stereo and its applications.	1	2	5
		OR			
3.	a).	Discuss Human Color Perception and the role of cones and rods	1	2	5
	b).	Compare Local Shading Models and Global Shading Models.	1	2	5
		UNIT-II			
4.	a).	Explain how filters are used as templates for pattern detection.	2	2	5
	b).	Discuss analysis and synthesis using oriented pyramids.	2	2	5
		OR			
5.	a).	Explain Spatial Frequency and Fourier Transforms in image processing.	2	2	5
	b).	Describe Shape from Texture and its applications.	2	2	5
		UNIT-III			

6.	a).	What is image segmentation? Explain its importance in image processing.	3	2	5
	b).	Explain different applications of segmentation in computer vision.	3	2	5
		OR			
7.	a).	Discuss the Gestalt principles used in human vision for grouping.	3	2	5
	b).	Discuss segmentation using graph-theoretic clustering methods.	3	2	5
		UNIT-IV			
8.	a).	Explain the concept of robustness in model fitting.	4	2	5
	b).	Explain Linear Dynamic Models used in object tracking.	4	2	5
		OR			
9.	a).	Describe the missing data problem in segmentation and fitting.	4	2	5
	b).	Discuss the Kalman Filtering algorithm with its applications.	4	2	5
		UNIT-V			
10.	a).	Explain the linear approach to camera calibration.	5	2	5
	b).	Explain analytical photogrammetry and its applications.	5	2	5
		OR			
11.	a).	Explain the linear approach to camera calibration.	5	2	5
	b).	Discuss the Least Squares Parameter Estimation method used in calibration.	5	2	5

CO-COURSE OUTCOME KL-KNOWLEDGE LEVEL M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23CB4104					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
DATA VISUALIZATION					
(for CSBS)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define data visualization. What is its primary objective?	1	1	2
	b).	State any two Gestalt principles used in visualization.	1	1	2
	c).	What is visual mapping in data visualization?	2	1	2
	d).	Define visual analytics.	2	1	2
	e).	What is meant by one-dimensional data visualization? Give one example.	3	1	2
	f).	What is interaction in visualization systems?	3	1	2
	g).	Define tree visualization	4	1	2
	h).	What is metaphorical visualization?	4	1	2
	i).	What is volumetric data visualization?	5	1	2
	j).	What is GIS in data visualization?	5	1	2
5 x 10 = 50 Marks					
		UNIT-1	CO	KL	M
2.	a).	Explain the visualization process and discuss the relationship between visualization and other fields.	1	2	5
	b).	Discuss visual perception in data visualization and explain the role of Gestalt principles with suitable examples.	1	3	5
		OR			
3.	a).	Define visualization and explain the history of visualization.	1	2	5
	b).	Explain information overload in visualization and discuss methods to overcome it.	1	3	5
		UNIT-2			
4.	a).	Explain the process of creating visual representations in data visualization.	2	2	5
	b).	Discuss the visualization reference model and explain its components.	2	3	5
		OR			
5.	a).	Explain visual mapping and its significance in visualization design.	2	2	5

	b).	Discuss the design principles of visualization applications with suitable examples.	2	3	5
		UNIT-3			
6.	a).	Classify visualization systems and explain their importance in data representation.	3	2	5
	b).	Explain interaction techniques in visualization systems with suitable examples	3	3	5
		OR			
7.	a).	Explain the visualization of one-dimensional, two-dimensional, and multi-dimensional data.	3	2	5
	b).	Discuss misleading visualization techniques and explain how they can affect interpretation of data.	3	3	5
		UNIT-4			
8.	a).	Explain the visualization of groups and trees with suitable examples.	4	2	5
	b).	Discuss graph and network visualization techniques used in data analysis.	4	3	5
		OR			
9.	a).	Explain the role of visualization software in creating charts and graphical representations.	4	2	5
	b).	Discuss metaphorical visualization and its applications in data visualization.	4	3	5
		UNIT-5			
10.	a).	Explain volumetric data visualization and discuss its applications.	5	2	5
	b).	Discuss GIS systems and map-based visualization in real-world applications.	5	2	5
		OR			
11.	a).	Explain visualization of vector fields, processes, and simulations with examples.	5	2	5
	b).	Discuss recent trends in visualization techniques and evaluation of visualizations.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks